Thaddeus Harris | Materials for the built environment | Homework #2 | 9/11/2024

(2) A 1 inch by 1 inch square steel bar is 12 inches long. The bar is loaded in tension so that it experiences an axial strain of 0.5%.

(a) At this strain, what are the new dimensions & cross sectional area of the bar if Poisson's ratio is 0.3?

$\varepsilon_x = 0.005 = \frac{\Delta L}{12}$

$\Delta L = 0.005 \cdot 12$

$\Delta L = 0.06$

$L_x = \Delta L_x + L_{ox}$

$L_x = 0.06 + 12$

$L_x = 12.06$

$r = 0.3 = -\frac{\varepsilon_y}{\varepsilon_x}$

$0.3 = -\frac{\varepsilon_y}{0.005}$

$-\varepsilon_y = 0.0015$

$\varepsilon_y = -0.0015$

$\Delta L_y = \varepsilon_y \cdot L_{oy}$

$\Delta L_y = -0.0015 \cdot 1$

$\Delta L_y = -0.0015$

$L_y = \Delta L_y + L_{oy}$

$L_y = -0.0015 + 1 = 0.9985$

$$\boxed{\begin{aligned} L_x &= 12.06 \\ L_y &= 0.9985 \end{aligned}}$$

(b) E = 29000 ksi for the steel. Based on the original bar dimensions and 0.5 percent axial strain what tensile stress exists in the bar assuming linear elastic behavior?

$E = \frac{\Delta\sigma}{\Delta\varepsilon} = 29000 \text{ ksi}$

$\frac{\Delta\sigma}{0.005} = 29000 \text{ ksi}$

$\Delta\sigma = 29000 \cdot 0.005$

$\sigma = 145 \text{ ksi}$

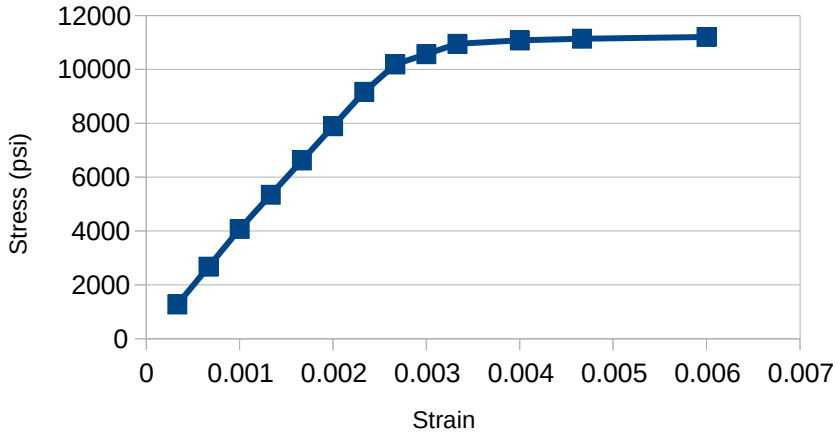
(c) What tensile force is required to produce the stress calculated in part (b)

$\sigma = 145 \text{ ksi} = \frac{F}{A}$

$145 \text{ ksi} = \frac{F}{1 \text{ in} \cdot 1 \text{ in}}$

$145 \text{ N/in}^2 = F/\text{in}^2$

$F = 145 \text{ klb}$ lbs

Load (lb)	Elongation (in)	Stress (psi)	Strain
0	0	0	0
40	0.001	1273.239545	0.000333333
84	0.002	2673.803044	0.000666667
128	0.003	4074.366543	0.001
168	0.004	5347.606088	0.001333333
208	0.005	6620.845633	0.001666667
248	0.006	7894.085177	0.002
288	0.007	9167.324722	0.002333333
320	0.008	10185.91636	0.002666667
332	0.009	10567.88822	0.003
344	0.01	10949.86008	0.003333333
348	0.012	11077.18404	0.004
350	0.014	11140.84602	0.004666667
352	0.018	11204.50799	0.006

(c) Stress-Strain Graph